

Feed rate (mm/rev)	Feed rate (mm/rev)	Tool material	Coolant usage (L/min)	Workpiece hardness (HB)	Cut time (min)	Tool path length (mm)	Tool usage frequency (uses/day)	Depth of cut (mm)	Vibration level (mm/s)	Tool wear (mm)
0.1	1000	Carbide	1	180	2	200	5	0.5	0.3	0.01
0.15	1200	Carbide	1.5	190	3	250	6	0.6	0.35	0.015
0.2	1400	HSS	2	200	4	300	7	0.7	0.4	0.02
0.25	1600	HSS	2.5	210	5	350	8	0.8	0.45	0.025
0.3	1800	Carbide	3	220	6	400	9	0.9	0.5	0.03
0.35	2000	Carbide	3.5	230	7	450	10	1	0.55	0.035
0.4	2200	Carbide	4	240	8	500	11	1.1	0.6	0.04
0.45	2400	HSS	4.5	250	9	550	12	1.2	0.65	0.045
0.5	2600	HSS	5	260	10	600	13	1.3	0.7	0.05
0.55	2800	Carbide	5.5	270	11	650	14	1.4	0.75	0.055
0.6	3000	Carbide	6	280	12	700	15	1.5	0.8	0.06
0.65	3200	Carbide	6.5	290	13	750	16	1.6	0.85	0.065
0.7	3400	HSS	7	300	14	800	17	1.7	0.9	0.07
0.75	3600	HSS	7.5	310	15	850	18	1.8	0.95	0.075
0.8	3800	Carbide	8	320	16	900	19	1.9	1	0.08
0.85	4000	Carbide	8.5	330	17	950	20	2	1.05	0.085
0.9	4200	Carbide	9	340	18	1000	21	2.1	1.1	0.09
0.95	4400	HSS	9.5	350	19	1050	22	2.2	1.15	0.095
1	4600	HSS	10	360	20	1100	23	2.3	1.2	0.1
1.05	4800	Carbide	10.5	370	21	1150	24	2.4	1.25	0.105
1.1	5000	Carbide	11	380	22	1200	25	2.5	1.3	0.11
1.15	5200	Carbide	11.5	390	23	1250	26	2.6	1.35	0.115
1.2	5400	HSS	12	400	24	1300	27	2.7	1.4	0.12
1.25	5600	HSS	12.5	410	25	1350	28	2.8	1.45	0.125
1.3	5800	Carbide	13	420	26	1400	29	2.9	1.5	0.13
1.35	6000	Carbide	13.5	430	27	1450	30	3	1.55	0.135
1.4	6200	Carbide	14	440	28	1500	31	3.1	1.6	0.14
1.45	6400	HSS	14.5	450	29	1550	32	3.2	1.65	0.145
1.5	6600	HSS	15	460	30	1600	33	3.3	1.7	0.15
1.55	6800	Carbide	15.5	470	31	1650	34	3.4	1.75	0.155
1.6	7000	Carbide	16	480	32	1700	35	3.5	1.8	0.16
1.65	7200	Carbide	16.5	490	33	1750	36	3.6	1.85	0.165
1.7	7400	HSS	17	500	34	1800	37	3.7	1.9	0.17
1.75	7600	HSS	17.5	510	35	1850	38	3.8	1.95	0.175
1.8	7800	Carbide	18	520	36	1900	39	3.9	2	0.18
1.85	8000	Carbide	18.5	530	37	1950	40	4	2.05	0.185
1.9	8200	Carbide	19	540	38	2000	41	4.1	2.1	0.19
1.95	8400	HSS	19.5	550	39	2050	42	4.2	2.15	0.195
2	8600	HSS	20	560	40	2100	43	4.3	2.2	0.2
2.05	8800	Carbide	20.5	570	41	2150	44	4.4	2.25	0.205
2.1	9000	Carbide	21	580	42	2200	45	4.5	2.3	0.21
2.15	9200	Carbide	21.5	590	43	2250	46	4.6	2.35	0.215
2.2	9400	HSS	22	600	44	2300	47	4.7	2.4	0.22
2.25	9600	HSS	22.5	610	45	2350	48	4.8	2.45	0.225
2.3	9800	Carbide	23	620	46	2400	49	4.9	2.5	0.23
2.35	10000	Carbide	23.5	630	47	2450	50	5	2.55	0.235

2.4	10200 Carbide	24	640	48	2500	51	5.1	2.6	0.24
2.45	10400 HSS	24.5	650	49	2550	52	5.2	2.65	0.245
2.5	10600 HSS	25	660	50	2600	53	5.3	2.7	0.25
2.55	10800 Carbide	25.5	670	51	2650	54	5.4	2.75	0.255
2.6	11000 Carbide	26	680	52	2700	55	5.5	2.8	0.26
2.65	11200 Carbide	26.5	690	53	2750	56	5.6	2.85	0.265
2.7	11400 HSS	27	700	54	2800	57	5.7	2.9	0.27
2.75	11600 HSS	27.5	710	55	2850	58	5.8	2.95	0.275
2.8	11800 Carbide	28	720	56	2900	59	5.9	3	0.28
2.85	12000 Carbide	28.5	730	57	2950	60	6	3.05	0.285
2.9	12200 Carbide	29	740	58	3000	61	6.1	3.1	0.29
2.95	12400 HSS	29.5	750	59	3050	62	6.2	3.15	0.295
3	12600 HSS	30	760	60	3100	63	6.3	3.2	0.3
3.05	12800 Carbide	30.5	770	61	3150	64	6.4	3.25	0.305
3.1	13000 Carbide	31	780	62	3200	65	6.5	3.3	0.31
3.15	13200 Carbide	31.5	790	63	3250	66	6.6	3.35	0.315
3.2	13400 HSS	32	800	64	3300	67	6.7	3.4	0.32
3.25	13600 HSS	32.5	810	65	3350	68	6.8	3.45	0.325
3.3	13800 Carbide	33	820	66	3400	69	6.9	3.5	0.33
3.35	14000 Carbide	33.5	830	67	3450	70	7	3.55	0.335
3.4	14200 Carbide	34	840	68	3500	71	7.1	3.6	0.34
3.45	14400 HSS	34.5	850	69	3550	72	7.2	3.65	0.345
3.5	14600 HSS	35	860	70	3600	73	7.3	3.7	0.35
3.55	14800 Carbide	35.5	870	71	3650	74	7.4	3.75	0.355
3.6	15000 Carbide	36	880	72	3700	75	7.5	3.8	0.36
3.65	15200 Carbide	36.5	890	73	3750	76	7.6	3.85	0.365
3.7	15400 HSS	37	900	74	3800	77	7.7	3.9	0.37
3.75	15600 HSS	37.5	910	75	3850	78	7.8	3.95	0.375
3.8	15800 Carbide	38	920	76	3900	79	7.9	4	0.38
3.85	16000 Carbide	38.5	930	77	3950	80	8	4.05	0.385
3.9	16200 Carbide	39	940	78	4000	81	8.1	4.1	0.39
3.95	16400 HSS	39.5	950	79	4050	82	8.2	4.15	0.395
4	16600 HSS	40	960	80	4100	83	8.3	4.2	0.4
4.05	16800 Carbide	40.5	970	81	4150	84	8.4	4.25	0.405
4.1	17000 Carbide	41	980	82	4200	85	8.5	4.3	0.41
4.15	17200 Carbide	41.5	990	83	4250	86	8.6	4.35	0.415
4.2	17400 HSS	42	1000	84	4300	87	8.7	4.4	0.42
4.25	17600 HSS	42.5	1010	85	4350	88	8.8	4.45	0.425
4.3	17800 Carbide	43	1020	86	4400	89	8.9	4.5	0.43
4.35	18000 Carbide	43.5	1030	87	4450	90	9	4.55	0.435
4.4	18200 Carbide	44	1040	88	4500	91	9.1	4.6	0.44
4.45	18400 HSS	44.5	1050	89	4550	92	9.2	4.65	0.445
4.5	18600 HSS	45	1060	90	4600	93	9.3	4.7	0.45
4.55	18800 Carbide	45.5	1070	91	4650	94	9.4	4.75	0.455
4.6	19000 Carbide	46	1080	92	4700	95	9.5	4.8	0.46
4.65	19200 Carbide	46.5	1090	93	4750	96	9.6	4.85	0.465
4.7	19400 HSS	47	1100	94	4800	97	9.7	4.9	0.47

4.75	19600 HSS	47.5	1110	95	4850	98	9.8	4.95	0.475
4.8	19800 Carbide	48	1120	96	4900	99	9.9	5	0.48
4.85	20000 Carbide	48.5	1130	97	4950	100	10	5.05	0.485
4.9	20200 Carbide	49	1140	98	5000	101	10.1	5.1	0.49
4.95	20400 HSS	49.5	1150	99	5050	102	10.2	5.15	0.495
5	20600 HSS	50	1160	100	5100	103	10.3	5.2	0.5
5.05	20800 Carbide	50.5	1170	101	5150	104	10.4	5.25	0.505
5.1	21000 Carbide	51	1180	102	5200	105	10.5	5.3	0.51
5.15	21200 Carbide	51.5	1190	103	5250	106	10.6	5.35	0.515
5.2	21400 HSS	52	1200	104	5300	107	10.7	5.4	0.52
5.25	21600 HSS	52.5	1210	105	5350	108	10.8	5.45	0.525
5.3	21800 Carbide	53	1220	106	5400	109	10.9	5.5	0.53
5.35	22000 Carbide	53.5	1230	107	5450	110	11	5.55	0.535
5.4	22200 Carbide	54	1240	108	5500	111	11.1	5.6	0.54
5.45	22400 HSS	54.5	1250	109	5550	112	11.2	5.65	0.545
5.5	22600 HSS	55	1260	110	5600	113	11.3	5.7	0.55
5.55	22800 Carbide	55.5	1270	111	5650	114	11.4	5.75	0.555
5.6	23000 Carbide	56	1280	112	5700	115	11.5	5.8	0.56
5.65	23200 Carbide	56.5	1290	113	5750	116	11.6	5.85	0.565
5.7	23400 HSS	57	1300	114	5800	117	11.7	5.9	0.57
5.75	23600 HSS	57.5	1310	115	5850	118	11.8	5.95	0.575
5.8	23800 Carbide	58	1320	116	5900	119	11.9	6	0.58
5.85	24000 Carbide	58.5	1330	117	5950	120	12	6.05	0.585
5.9	24200 Carbide	59	1340	118	6000	121	12.1	6.1	0.59
5.95	24400 HSS	59.5	1350	119	6050	122	12.2	6.15	0.595
6	24600 HSS	60	1360	120	6100	123	12.3	6.2	0.6
6.05	24800 Carbide	60.5	1370	121	6150	124	12.4	6.25	0.605
6.1	25000 Carbide	61	1380	122	6200	125	12.5	6.3	0.61
6.15	25200 Carbide	61.5	1390	123	6250	126	12.6	6.35	0.615
6.2	25400 HSS	62	1400	124	6300	127	12.7	6.4	0.62
6.25	25600 HSS	62.5	1410	125	6350	128	12.8	6.45	0.625
6.3	25800 Carbide	63	1420	126	6400	129	12.9	6.5	0.63
6.35	26000 Carbide	63.5	1430	127	6450	130	13	6.55	0.635
6.4	26200 Carbide	64	1440	128	6500	131	13.1	6.6	0.64
6.45	26400 HSS	64.5	1450	129	6550	132	13.2	6.65	0.645
6.5	26600 HSS	65	1460	130	6600	133	13.3	6.7	0.65
6.55	26800 Carbide	65.5	1470	131	6650	134	13.4	6.75	0.655
6.6	27000 Carbide	66	1480	132	6700	135	13.5	6.8	0.66
6.65	27200 Carbide	66.5	1490	133	6750	136	13.6	6.85	0.665
6.7	27400 HSS	67	1500	134	6800	137	13.7	6.9	0.67
6.75	27600 HSS	67.5	1510	135	6850	138	13.8	6.95	0.675
6.8	27800 Carbide	68	1520	136	6900	139	13.9	7	0.68
6.85	28000 Carbide	68.5	1530	137	6950	140	14	7.05	0.685
6.9	28200 Carbide	69	1540	138	7000	141	14.1	7.1	0.69
6.95	28400 HSS	69.5	1550	139	7050	142	14.2	7.15	0.695
7	28600 HSS	70	1560	140	7100	143	14.3	7.2	0.7
7.05	28800 Carbide	70.5	1570	141	7150	144	14.4	7.25	0.705

7.1	29000 Carbide	71	1580	142	7200	145	14.5	7.3	0.71
7.15	29200 Carbide	71.5	1590	143	7250	146	14.6	7.35	0.715
7.2	29400 HSS	72	1600	144	7300	147	14.7	7.4	0.72
7.25	29600 HSS	72.5	1610	145	7350	148	14.8	7.45	0.725
7.3	29800 Carbide	73	1620	146	7400	149	14.9	7.5	0.73
7.35	30000 Carbide	73.5	1630	147	7450	150	15	7.55	0.735
7.4	30200 Carbide	74	1640	148	7500	151	15.1	7.6	0.74
7.45	30400 HSS	74.5	1650	149	7550	152	15.2	7.65	0.745
7.5	30600 HSS	75	1660	150	7600	153	15.3	7.7	0.75
7.55	30800 Carbide	75.5	1670	151	7650	154	15.4	7.75	0.755
7.6	31000 Carbide	76	1680	152	7700	155	15.5	7.8	0.76
7.65	31200 Carbide	76.5	1690	153	7750	156	15.6	7.85	0.765
7.7	31400 HSS	77	1700	154	7800	157	15.7	7.9	0.77
7.75	31600 HSS	77.5	1710	155	7850	158	15.8	7.95	0.775
7.8	31800 Carbide	78	1720	156	7900	159	15.9	8	0.78
7.85	32000 Carbide	78.5	1730	157	7950	160	16	8.05	0.785
7.9	32200 Carbide	79	1740	158	8000	161	16.1	8.1	0.79
7.95	32400 HSS	79.5	1750	159	8050	162	16.2	8.15	0.795
8	32600 HSS	80	1760	160	8100	163	16.3	8.2	0.8
8.05	32800 Carbide	80.5	1770	161	8150	164	16.4	8.25	0.805
8.1	33000 Carbide	81	1780	162	8200	165	16.5	8.3	0.81
8.15	33200 Carbide	81.5	1790	163	8250	166	16.6	8.35	0.815
8.2	33400 HSS	82	1800	164	8300	167	16.7	8.4	0.82
8.25	33600 HSS	82.5	1810	165	8350	168	16.8	8.45	0.825
8.3	33800 Carbide	83	1820	166	8400	169	16.9	8.5	0.83
8.35	34000 Carbide	83.5	1830	167	8450	170	17	8.55	0.835
8.4	34200 Carbide	84	1840	168	8500	171	17.1	8.6	0.84
8.45	34400 HSS	84.5	1850	169	8550	172	17.2	8.65	0.845
8.5	34600 HSS	85	1860	170	8600	173	17.3	8.7	0.85
8.55	34800 Carbide	85.5	1870	171	8650	174	17.4	8.75	0.855
8.6	35000 Carbide	86	1880	172	8700	175	17.5	8.8	0.86
8.65	35200 Carbide	86.5	1890	173	8750	176	17.6	8.85	0.865
8.7	35400 HSS	87	1900	174	8800	177	17.7	8.9	0.87
8.75	35600 HSS	87.5	1910	175	8850	178	17.8	8.95	0.875
8.8	35800 Carbide	88	1920	176	8900	179	17.9	9	0.88
8.85	36000 Carbide	88.5	1930	177	8950	180	18	9.05	0.885
8.9	36200 Carbide	89	1940	178	9000	181	18.1	9.1	0.89
8.95	36400 HSS	89.5	1950	179	9050	182	18.2	9.15	0.895
9	36600 HSS	90	1960	180	9100	183	18.3	9.2	0.9
9.05	36800 Carbide	90.5	1970	181	9150	184	18.4	9.25	0.905
9.1	37000 Carbide	91	1980	182	9200	185	18.5	9.3	0.91
9.15	37200 Carbide	91.5	1990	183	9250	186	18.6	9.35	0.915
9.2	37400 HSS	92	2000	184	9300	187	18.7	9.4	0.92
9.25	37600 HSS	92.5	2010	185	9350	188	18.8	9.45	0.925
9.3	37800 Carbide	93	2020	186	9400	189	18.9	9.5	0.93
9.35	38000 Carbide	93.5	2030	187	9450	190	19	9.55	0.935
9.4	38200 Carbide	94	2040	188	9500	191	19.1	9.6	0.94

9.45	38400 HSS	94.5	2050	189	9550	192	19.2	9.65	0.945
9.5	38600 HSS	95	2060	190	9600	193	19.3	9.7	0.95
9.55	38800 Carbide	95.5	2070	191	9650	194	19.4	9.75	0.955
9.6	39000 Carbide	96	2080	192	9700	195	19.5	9.8	0.96
9.65	39200 Carbide	96.5	2090	193	9750	196	19.6	9.85	0.965
9.7	39400 HSS	97	2100	194	9800	197	19.7	9.9	0.97
9.75	39600 HSS	97.5	2110	195	9850	198	19.8	9.95	0.975
9.8	39800 Carbide	98	2120	196	9900	199	19.9	10	0.98
9.85	40000 Carbide	98.5	2130	197	9950	200	20	10.05	0.985
9.9	40200 Carbide	99	2140	198	10000	201	20.1	10.1	0.99
9.95	40400 HSS	99.5	2150	199	10050	202	20.2	10.15	0.995
10	40600 HSS	100	2160	200	10100	203	20.3	10.2	1
10.05	40800 Carbide	100.5	2170	201	10150	204	20.4	10.25	1.005
10.1	41000 Carbide	101	2180	202	10200	205	20.5	10.3	1.01
10.15	41200 Carbide	101.5	2190	203	10250	206	20.6	10.35	1.015
10.2	41400 HSS	102	2200	204	10300	207	20.7	10.4	1.02
10.25	41600 HSS	102.5	2210	205	10350	208	20.8	10.45	1.025
10.3	41800 Carbide	103	2220	206	10400	209	20.9	10.5	1.03
10.35	42000 Carbide	103.5	2230	207	10450	210	21	10.55	1.035
10.4	42200 Carbide	104	2240	208	10500	211	21.1	10.6	1.04
10.45	42400 HSS	104.5	2250	209	10550	212	21.2	10.65	1.045
10.5	42600 HSS	105	2260	210	10600	213	21.3	10.7	1.05
10.55	42800 Carbide	105.5	2270	211	10650	214	21.4	10.75	1.055
10.6	43000 Carbide	106	2280	212	10700	215	21.5	10.8	1.06
10.65	43200 Carbide	106.5	2290	213	10750	216	21.6	10.85	1.065
10.7	43400 HSS	107	2300	214	10800	217	21.7	10.9	1.07
10.75	43600 HSS	107.5	2310	215	10850	218	21.8	10.95	1.075
10.8	43800 Carbide	108	2320	216	10900	219	21.9	11	1.08
10.85	44000 Carbide	108.5	2330	217	10950	220	22	11.05	1.085
10.9	44200 Carbide	109	2340	218	11000	221	22.1	11.1	1.09
10.95	44400 HSS	109.5	2350	219	11050	222	22.2	11.15	1.095
11	44600 HSS	110	2360	220	11100	223	22.3	11.2	1.1
11.05	44800 Carbide	110.5	2370	221	11150	224	22.4	11.25	1.105
11.1	45000 Carbide	111	2380	222	11200	225	22.5	11.3	1.11
11.15	45200 Carbide	111.5	2390	223	11250	226	22.6	11.35	1.115
11.2	45400 HSS	112	2400	224	11300	227	22.7	11.4	1.12
11.25	45600 HSS	112.5	2410	225	11350	228	22.8	11.45	1.125
11.3	45800 Carbide	113	2420	226	11400	229	22.9	11.5	1.13
11.35	46000 Carbide	113.5	2430	227	11450	230	23	11.55	1.135
11.4	46200 Carbide	114	2440	228	11500	231	23.1	11.6	1.14
11.45	46400 HSS	114.5	2450	229	11550	232	23.2	11.65	1.145
11.5	46600 HSS	115	2460	230	11600	233	23.3	11.7	1.15
11.55	46800 Carbide	115.5	2470	231	11650	234	23.4	11.75	1.155
11.6	47000 Carbide	116	2480	232	11700	235	23.5	11.8	1.16
11.65	47200 Carbide	116.5	2490	233	11750	236	23.6	11.85	1.165
11.7	47400 HSS	117	2500	234	11800	237	23.7	11.9	1.17
11.75	47600 HSS	117.5	2510	235	11850	238	23.8	11.95	1.175

11.8	47800 Carbide	118	2520	236	11900	239	23.9	12	1.18
11.85	48000 Carbide	118.5	2530	237	11950	240	24	12.05	1.185
11.9	48200 Carbide	119	2540	238	12000	241	24.1	12.1	1.19
11.95	48400 HSS	119.5	2550	239	12050	242	24.2	12.15	1.195
12	48600 HSS	120	2560	240	12100	243	24.3	12.2	1.2
12.05	48800 Carbide	120.5	2570	241	12150	244	24.4	12.25	1.205
12.1	49000 Carbide	121	2580	242	12200	245	24.5	12.3	1.21
12.15	49200 Carbide	121.5	2590	243	12250	246	24.6	12.35	1.215
12.2	49400 HSS	122	2600	244	12300	247	24.7	12.4	1.22
12.25	49600 HSS	122.5	2610	245	12350	248	24.8	12.45	1.225
12.3	49800 Carbide	123	2620	246	12400	249	24.9	12.5	1.23
12.35	50000 Carbide	123.5	2630	247	12450	250	25	12.55	1.235
12.4	50200 Carbide	124	2640	248	12500	251	25.1	12.6	1.24
12.45	50400 HSS	124.5	2650	249	12550	252	25.2	12.65	1.245
12.5	50600 HSS	125	2660	250	12600	253	25.3	12.7	1.25
12.55	50800 Carbide	125.5	2670	251	12650	254	25.4	12.75	1.255
12.6	51000 Carbide	126	2680	252	12700	255	25.5	12.8	1.26
12.65	51200 Carbide	126.5	2690	253	12750	256	25.6	12.85	1.265
12.7	51400 HSS	127	2700	254	12800	257	25.7	12.9	1.27
12.75	51600 HSS	127.5	2710	255	12850	258	25.8	12.95	1.275
12.8	51800 Carbide	128	2720	256	12900	259	25.9	13	1.28
12.85	52000 Carbide	128.5	2730	257	12950	260	26	13.05	1.285
12.9	52200 Carbide	129	2740	258	13000	261	26.1	13.1	1.29
12.95	52400 HSS	129.5	2750	259	13050	262	26.2	13.15	1.295
13	52600 HSS	130	2760	260	13100	263	26.3	13.2	1.3
13.05	52800 Carbide	130.5	2770	261	13150	264	26.4	13.25	1.305
13.1	53000 Carbide	131	2780	262	13200	265	26.5	13.3	1.31
13.15	53200 Carbide	131.5	2790	263	13250	266	26.6	13.35	1.315
13.2	53400 HSS	132	2800	264	13300	267	26.7	13.4	1.32
13.25	53600 HSS	132.5	2810	265	13350	268	26.8	13.45	1.325
13.3	53800 Carbide	133	2820	266	13400	269	26.9	13.5	1.33
13.35	54000 Carbide	133.5	2830	267	13450	270	27	13.55	1.335
13.4	54200 Carbide	134	2840	268	13500	271	27.1	13.6	1.34
13.45	54400 HSS	134.5	2850	269	13550	272	27.2	13.65	1.345
13.5	54600 HSS	135	2860	270	13600	273	27.3	13.7	1.35
13.55	54800 Carbide	135.5	2870	271	13650	274	27.4	13.75	1.355
13.6	55000 Carbide	136	2880	272	13700	275	27.5	13.8	1.36
13.65	55200 Carbide	136.5	2890	273	13750	276	27.6	13.85	1.365
13.7	55400 HSS	137	2900	274	13800	277	27.7	13.9	1.37
13.75	55600 HSS	137.5	2910	275	13850	278	27.8	13.95	1.375
13.8	55800 Carbide	138	2920	276	13900	279	27.9	14	1.38
13.85	56000 Carbide	138.5	2930	277	13950	280	28	14.05	1.385
13.9	56200 Carbide	139	2940	278	14000	281	28.1	14.1	1.39
13.95	56400 HSS	139.5	2950	279	14050	282	28.2	14.15	1.395
14	56600 HSS	140	2960	280	14100	283	28.3	14.2	1.4
14.05	56800 Carbide	140.5	2970	281	14150	284	28.4	14.25	1.405
14.1	57000 Carbide	141	2980	282	14200	285	28.5	14.3	1.41

14.15	57200 Carbide	141.5	2990	283	14250	286	28.6	14.35	1.415
14.2	57400 HSS	142	3000	284	14300	287	28.7	14.4	1.42
14.25	57600 HSS	142.5	3010	285	14350	288	28.8	14.45	1.425
14.3	57800 Carbide	143	3020	286	14400	289	28.9	14.5	1.43
14.35	58000 Carbide	143.5	3030	287	14450	290	29	14.55	1.435
14.4	58200 Carbide	144	3040	288	14500	291	29.1	14.6	1.44
14.45	58400 HSS	144.5	3050	289	14550	292	29.2	14.65	1.445
14.5	58600 HSS	145	3060	290	14600	293	29.3	14.7	1.45
14.55	58800 Carbide	145.5	3070	291	14650	294	29.4	14.75	1.455
14.6	59000 Carbide	146	3080	292	14700	295	29.5	14.8	1.46
14.65	59200 Carbide	146.5	3090	293	14750	296	29.6	14.85	1.465
14.7	59400 HSS	147	3100	294	14800	297	29.7	14.9	1.47
14.75	59600 HSS	147.5	3110	295	14850	298	29.8	14.95	1.475
14.8	59800 Carbide	148	3120	296	14900	299	29.9	15	1.48
14.85	60000 Carbide	148.5	3130	297	14950	300	30	15.05	1.485
14.9	60200 Carbide	149	3140	298	15000	301	30.1	15.1	1.49
14.95	60400 HSS	149.5	3150	299	15050	302	30.2	15.15	1.495
15	60600 HSS	150	3160	300	15100	303	30.3	15.2	1.5
15.05	60800 Carbide	150.5	3170	301	15150	304	30.4	15.25	1.505
15.1	61000 Carbide	151	3180	302	15200	305	30.5	15.3	1.51
15.15	61200 Carbide	151.5	3190	303	15250	306	30.6	15.35	1.515
15.2	61400 HSS	152	3200	304	15300	307	30.7	15.4	1.52
15.25	61600 HSS	152.5	3210	305	15350	308	30.8	15.45	1.525
15.3	61800 Carbide	153	3220	306	15400	309	30.9	15.5	1.53
15.35	62000 Carbide	153.5	3230	307	15450	310	31	15.55	1.535
15.4	62200 Carbide	154	3240	308	15500	311	31.1	15.6	1.54
15.45	62400 HSS	154.5	3250	309	15550	312	31.2	15.65	1.545
15.5	62600 HSS	155	3260	310	15600	313	31.3	15.7	1.55
15.55	62800 Carbide	155.5	3270	311	15650	314	31.4	15.75	1.555
15.6	63000 Carbide	156	3280	312	15700	315	31.5	15.8	1.56
15.65	63200 Carbide	156.5	3290	313	15750	316	31.6	15.85	1.565
15.7	63400 HSS	157	3300	314	15800	317	31.7	15.9	1.57
15.75	63600 HSS	157.5	3310	315	15850	318	31.8	15.95	1.575
15.8	63800 Carbide	158	3320	316	15900	319	31.9	16	1.58
15.85	64000 Carbide	158.5	3330	317	15950	320	32	16.05	1.585
15.9	64200 Carbide	159	3340	318	16000	321	32.1	16.1	1.59
15.95	64400 HSS	159.5	3350	319	16050	322	32.2	16.15	1.595
16	64600 HSS	160	3360	320	16100	323	32.3	16.2	1.6
16.05	64800 Carbide	160.5	3370	321	16150	324	32.4	16.25	1.605
16.1	65000 Carbide	161	3380	322	16200	325	32.5	16.3	1.61
16.15	65200 Carbide	161.5	3390	323	16250	326	32.6	16.35	1.615
16.2	65400 HSS	162	3400	324	16300	327	32.7	16.4	1.62
16.25	65600 HSS	162.5	3410	325	16350	328	32.8	16.45	1.625
16.3	65800 Carbide	163	3420	326	16400	329	32.9	16.5	1.63
16.35	66000 Carbide	163.5	3430	327	16450	330	33	16.55	1.635
16.4	66200 Carbide	164	3440	328	16500	331	33.1	16.6	1.64
16.45	66400 HSS	164.5	3450	329	16550	332	33.2	16.65	1.645

16.5	66600 HSS	165	3460	330	16600	333	33.3	16.7	1.65
16.55	66800 Carbide	165.5	3470	331	16650	334	33.4	16.75	1.655
16.6	67000 Carbide	166	3480	332	16700	335	33.5	16.8	1.66
16.65	67200 Carbide	166.5	3490	333	16750	336	33.6	16.85	1.665
16.7	67400 HSS	167	3500	334	16800	337	33.7	16.9	1.67
16.75	67600 HSS	167.5	3510	335	16850	338	33.8	16.95	1.675
16.8	67800 Carbide	168	3520	336	16900	339	33.9	17	1.68
16.85	68000 Carbide	168.5	3530	337	16950	340	34	17.05	1.685
16.9	68200 Carbide	169	3540	338	17000	341	34.1	17.1	1.69
16.95	68400 HSS	169.5	3550	339	17050	342	34.2	17.15	1.695
17	68600 HSS	170	3560	340	17100	343	34.3	17.2	1.7
17.05	68800 Carbide	170.5	3570	341	17150	344	34.4	17.25	1.705
17.1	69000 Carbide	171	3580	342	17200	345	34.5	17.3	1.71
17.15	69200 Carbide	171.5	3590	343	17250	346	34.6	17.35	1.715
17.2	69400 HSS	172	3600	344	17300	347	34.7	17.4	1.72
17.25	69600 HSS	172.5	3610	345	17350	348	34.8	17.45	1.725
17.3	69800 Carbide	173	3620	346	17400	349	34.9	17.5	1.73
17.35	70000 Carbide	173.5	3630	347	17450	350	35	17.55	1.735
17.4	70200 Carbide	174	3640	348	17500	351	35.1	17.6	1.74
17.45	70400 HSS	174.5	3650	349	17550	352	35.2	17.65	1.745
17.5	70600 HSS	175	3660	350	17600	353	35.3	17.7	1.75
17.55	70800 Carbide	175.5	3670	351	17650	354	35.4	17.75	1.755
17.6	71000 Carbide	176	3680	352	17700	355	35.5	17.8	1.76
17.65	71200 Carbide	176.5	3690	353	17750	356	35.6	17.85	1.765
17.7	71400 HSS	177	3700	354	17800	357	35.7	17.9	1.77
17.75	71600 HSS	177.5	3710	355	17850	358	35.8	17.95	1.775
17.8	71800 Carbide	178	3720	356	17900	359	35.9	18	1.78
17.85	72000 Carbide	178.5	3730	357	17950	360	36	18.05	1.785
17.9	72200 Carbide	179	3740	358	18000	361	36.1	18.1	1.79
17.95	72400 HSS	179.5	3750	359	18050	362	36.2	18.15	1.795
18	72600 HSS	180	3760	360	18100	363	36.3	18.2	1.8
18.05	72800 Carbide	180.5	3770	361	18150	364	36.4	18.25	1.805
18.1	73000 Carbide	181	3780	362	18200	365	36.5	18.3	1.81
18.15	73200 Carbide	181.5	3790	363	18250	366	36.6	18.35	1.815
18.2	73400 HSS	182	3800	364	18300	367	36.7	18.4	1.82
18.25	73600 HSS	182.5	3810	365	18350	368	36.8	18.45	1.825
18.3	73800 Carbide	183	3820	366	18400	369	36.9	18.5	1.83
18.35	74000 Carbide	183.5	3830	367	18450	370	37	18.55	1.835
18.4	74200 Carbide	184	3840	368	18500	371	37.1	18.6	1.84
18.45	74400 HSS	184.5	3850	369	18550	372	37.2	18.65	1.845
18.5	74600 HSS	185	3860	370	18600	373	37.3	18.7	1.85
18.55	74800 Carbide	185.5	3870	371	18650	374	37.4	18.75	1.855
18.6	75000 Carbide	186	3880	372	18700	375	37.5	18.8	1.86
18.65	75200 Carbide	186.5	3890	373	18750	376	37.6	18.85	1.865
18.7	75400 HSS	187	3900	374	18800	377	37.7	18.9	1.87
18.75	75600 HSS	187.5	3910	375	18850	378	37.8	18.95	1.875
18.8	75800 Carbide	188	3920	376	18900	379	37.9	19	1.88

18.85	76000 Carbide	188.5	3930	377	18950	380	38	19.05	1.885
18.9	76200 Carbide	189	3940	378	19000	381	38.1	19.1	1.89
18.95	76400 HSS	189.5	3950	379	19050	382	38.2	19.15	1.895
19	76600 HSS	190	3960	380	19100	383	38.3	19.2	1.9
19.05	76800 Carbide	190.5	3970	381	19150	384	38.4	19.25	1.905
19.1	77000 Carbide	191	3980	382	19200	385	38.5	19.3	1.91
19.15	77200 Carbide	191.5	3990	383	19250	386	38.6	19.35	1.915
19.2	77400 HSS	192	4000	384	19300	387	38.7	19.4	1.92
19.25	77600 HSS	192.5	4010	385	19350	388	38.8	19.45	1.925
19.3	77800 Carbide	193	4020	386	19400	389	38.9	19.5	1.93
19.35	78000 Carbide	193.5	4030	387	19450	390	39	19.55	1.935
19.4	78200 Carbide	194	4040	388	19500	391	39.1	19.6	1.94
19.45	78400 HSS	194.5	4050	389	19550	392	39.2	19.65	1.945
19.5	78600 HSS	195	4060	390	19600	393	39.3	19.7	1.95
19.55	78800 Carbide	195.5	4070	391	19650	394	39.4	19.75	1.955
19.6	79000 Carbide	196	4080	392	19700	395	39.5	19.8	1.96
19.65	79200 Carbide	196.5	4090	393	19750	396	39.6	19.85	1.965
19.7	79400 HSS	197	4100	394	19800	397	39.7	19.9	1.97
19.75	79600 HSS	197.5	4110	395	19850	398	39.8	19.95	1.975
19.8	79800 Carbide	198	4120	396	19900	399	39.9	20	1.98
19.85	80000 Carbide	198.5	4130	397	19950	400	40	20.05	1.985
19.9	80200 Carbide	199	4140	398	20000	401	40.1	20.1	1.99
19.95	80400 HSS	199.5	4150	399	20050	402	40.2	20.15	1.995
20	80600 HSS	200	4160	400	20100	403	40.3	20.2	2
20.05	80800 Carbide	200.5	4170	401	20150	404	40.4	20.25	2.005
20.1	81000 Carbide	201	4180	402	20200	405	40.5	20.3	2.01
20.15	81200 Carbide	201.5	4190	403	20250	406	40.6	20.35	2.015
20.2	81400 HSS	202	4200	404	20300	407	40.7	20.4	2.02
20.25	81600 HSS	202.5	4210	405	20350	408	40.8	20.45	2.025
20.3	81800 Carbide	203	4220	406	20400	409	40.9	20.5	2.03
20.35	82000 Carbide	203.5	4230	407	20450	410	41	20.55	2.035
20.4	82200 Carbide	204	4240	408	20500	411	41.1	20.6	2.04
20.45	82400 HSS	204.5	4250	409	20550	412	41.2	20.65	2.045
20.5	82600 HSS	205	4260	410	20600	413	41.3	20.7	2.05
20.55	82800 Carbide	205.5	4270	411	20650	414	41.4	20.75	2.055
20.6	83000 Carbide	206	4280	412	20700	415	41.5	20.8	2.06
20.65	83200 Carbide	206.5	4290	413	20750	416	41.6	20.85	2.065
20.7	83400 HSS	207	4300	414	20800	417	41.7	20.9	2.07
20.75	83600 HSS	207.5	4310	415	20850	418	41.8	20.95	2.075
20.8	83800 Carbide	208	4320	416	20900	419	41.9	21	2.08
20.85	84000 Carbide	208.5	4330	417	20950	420	42	21.05	2.085
20.9	84200 Carbide	209	4340	418	21000	421	42.1	21.1	2.09
20.95	84400 HSS	209.5	4350	419	21050	422	42.2	21.15	2.095
21	84600 HSS	210	4360	420	21100	423	42.3	21.2	2.1
21.05	84800 Carbide	210.5	4370	421	21150	424	42.4	21.25	2.105
21.1	85000 Carbide	211	4380	422	21200	425	42.5	21.3	2.11
21.15	85200 Carbide	211.5	4390	423	21250	426	42.6	21.35	2.115

21.2	85400 HSS	212	4400	424	21300	427	42.7	21.4	2.12
21.25	85600 HSS	212.5	4410	425	21350	428	42.8	21.45	2.125
21.3	85800 Carbide	213	4420	426	21400	429	42.9	21.5	2.13
21.35	86000 Carbide	213.5	4430	427	21450	430	43	21.55	2.135
21.4	86200 Carbide	214	4440	428	21500	431	43.1	21.6	2.14
21.45	86400 HSS	214.5	4450	429	21550	432	43.2	21.65	2.145
21.5	86600 HSS	215	4460	430	21600	433	43.3	21.7	2.15
21.55	86800 Carbide	215.5	4470	431	21650	434	43.4	21.75	2.155
21.6	87000 Carbide	216	4480	432	21700	435	43.5	21.8	2.16
21.65	87200 Carbide	216.5	4490	433	21750	436	43.6	21.85	2.165
21.7	87400 HSS	217	4500	434	21800	437	43.7	21.9	2.17
21.75	87600 HSS	217.5	4510	435	21850	438	43.8	21.95	2.175
21.8	87800 Carbide	218	4520	436	21900	439	43.9	22	2.18
21.85	88000 Carbide	218.5	4530	437	21950	440	44	22.05	2.185
21.9	88200 Carbide	219	4540	438	22000	441	44.1	22.1	2.19
21.95	88400 HSS	219.5	4550	439	22050	442	44.2	22.15	2.195
22	88600 HSS	220	4560	440	22100	443	44.3	22.2	2.2
22.05	88800 Carbide	220.5	4570	441	22150	444	44.4	22.25	2.205
22.1	89000 Carbide	221	4580	442	22200	445	44.5	22.3	2.21
22.15	89200 Carbide	221.5	4590	443	22250	446	44.6	22.35	2.215
22.2	89400 HSS	222	4600	444	22300	447	44.7	22.4	2.22
22.25	89600 HSS	222.5	4610	445	22350	448	44.8	22.45	2.225
22.3	89800 Carbide	223	4620	446	22400	449	44.9	22.5	2.23
22.35	90000 Carbide	223.5	4630	447	22450	450	45	22.55	2.235
22.4	90200 Carbide	224	4640	448	22500	451	45.1	22.6	2.24
22.45	90400 HSS	224.5	4650	449	22550	452	45.2	22.65	2.245
22.5	90600 HSS	225	4660	450	22600	453	45.3	22.7	2.25
22.55	90800 Carbide	225.5	4670	451	22650	454	45.4	22.75	2.255
22.6	91000 Carbide	226	4680	452	22700	455	45.5	22.8	2.26
22.65	91200 Carbide	226.5	4690	453	22750	456	45.6	22.85	2.265
22.7	91400 HSS	227	4700	454	22800	457	45.7	22.9	2.27
22.75	91600 HSS	227.5	4710	455	22850	458	45.8	22.95	2.275
22.8	91800 Carbide	228	4720	456	22900	459	45.9	23	2.28
22.85	92000 Carbide	228.5	4730	457	22950	460	46	23.05	2.285
22.9	92200 Carbide	229	4740	458	23000	461	46.1	23.1	2.29
22.95	92400 HSS	229.5	4750	459	23050	462	46.2	23.15	2.295
23	92600 HSS	230	4760	460	23100	463	46.3	23.2	2.3
23.05	92800 Carbide	230.5	4770	461	23150	464	46.4	23.25	2.305
23.1	93000 Carbide	231	4780	462	23200	465	46.5	23.3	2.31
23.15	93200 Carbide	231.5	4790	463	23250	466	46.6	23.35	2.315
23.2	93400 HSS	232	4800	464	23300	467	46.7	23.4	2.32
23.25	93600 HSS	232.5	4810	465	23350	468	46.8	23.45	2.325
23.3	93800 Carbide	233	4820	466	23400	469	46.9	23.5	2.33
23.35	94000 Carbide	233.5	4830	467	23450	470	47	23.55	2.335
23.4	94200 Carbide	234	4840	468	23500	471	47.1	23.6	2.34
23.45	94400 HSS	234.5	4850	469	23550	472	47.2	23.65	2.345
23.5	94600 HSS	235	4860	470	23600	473	47.3	23.7	2.35

23.55	94800 Carbide	235.5	4870	471	23650	474	47.4	23.75	2.355
23.6	95000 Carbide	236	4880	472	23700	475	47.5	23.8	2.36
23.65	95200 Carbide	236.5	4890	473	23750	476	47.6	23.85	2.365
23.7	95400 HSS	237	4900	474	23800	477	47.7	23.9	2.37
23.75	95600 HSS	237.5	4910	475	23850	478	47.8	23.95	2.375
23.8	95800 Carbide	238	4920	476	23900	479	47.9	24	2.38
23.85	96000 Carbide	238.5	4930	477	23950	480	48	24.05	2.385
23.9	96200 Carbide	239	4940	478	24000	481	48.1	24.1	2.39
23.95	96400 HSS	239.5	4950	479	24050	482	48.2	24.15	2.395
24	96600 HSS	240	4960	480	24100	483	48.3	24.2	2.4
24.05	96800 Carbide	240.5	4970	481	24150	484	48.4	24.25	2.405
24.1	97000 Carbide	241	4980	482	24200	485	48.5	24.3	2.41
24.15	97200 Carbide	241.5	4990	483	24250	486	48.6	24.35	2.415
24.2	97400 HSS	242	5000	484	24300	487	48.7	24.4	2.42
24.25	97600 HSS	242.5	5010	485	24350	488	48.8	24.45	2.425
24.3	97800 Carbide	243	5020	486	24400	489	48.9	24.5	2.43
24.35	98000 Carbide	243.5	5030	487	24450	490	49	24.55	2.435
24.4	98200 Carbide	244	5040	488	24500	491	49.1	24.6	2.44
24.45	98400 HSS	244.5	5050	489	24550	492	49.2	24.65	2.445
24.5	98600 HSS	245	5060	490	24600	493	49.3	24.7	2.45
24.55	98800 Carbide	245.5	5070	491	24650	494	49.4	24.75	2.455
24.6	99000 Carbide	246	5080	492	24700	495	49.5	24.8	2.46
24.65	99200 Carbide	246.5	5090	493	24750	496	49.6	24.85	2.465
24.7	99400 HSS	247	5100	494	24800	497	49.7	24.9	2.47
24.75	99600 HSS	247.5	5110	495	24850	498	49.8	24.95	2.475
24.8	99800 Carbide	248	5120	496	24900	499	49.9	25	2.48
24.85	100000 Carbide	248.5	5130	497	24950	500	50	25.05	2.485
24.9	100200 Carbide	249	5140	498	25000	501	50.1	25.1	2.49
24.95	100400 HSS	249.5	5150	499	25050	502	50.2	25.15	2.495
25	100600 HSS	250	5160	500	25100	503	50.3	25.2	2.5
25.05	100800 Carbide	250.5	5170	501	25150	504	50.4	25.25	2.505
25.1	101000 Carbide	251	5180	502	25200	505	50.5	25.3	2.51
25.15	101200 Carbide	251.5	5190	503	25250	506	50.6	25.35	2.515
25.2	101400 HSS	252	5200	504	25300	507	50.7	25.4	2.52
25.25	101600 HSS	252.5	5210	505	25350	508	50.8	25.45	2.525
25.3	101800 Carbide	253	5220	506	25400	509	50.9	25.5	2.53
25.35	102000 Carbide	253.5	5230	507	25450	510	51	25.55	2.535
25.4	102200 Carbide	254	5240	508	25500	511	51.1	25.6	2.54
25.45	102400 HSS	254.5	5250	509	25550	512	51.2	25.65	2.545
25.5	102600 HSS	255	5260	510	25600	513	51.3	25.7	2.55
25.55	102800 Carbide	255.5	5270	511	25650	514	51.4	25.75	2.555
25.6	103000 Carbide	256	5280	512	25700	515	51.5	25.8	2.56
25.65	103200 Carbide	256.5	5290	513	25750	516	51.6	25.85	2.565
25.7	103400 HSS	257	5300	514	25800	517	51.7	25.9	2.57
25.75	103600 HSS	257.5	5310	515	25850	518	51.8	25.95	2.575
25.8	103800 Carbide	258	5320	516	25900	519	51.9	26	2.58
25.85	104000 Carbide	258.5	5330	517	25950	520	52	26.05	2.585

25.9	104200 Carbide	259	5340	518	26000	521	52.1	26.1	2.59
25.95	104400 HSS	259.5	5350	519	26050	522	52.2	26.15	2.595
26	104600 HSS	260	5360	520	26100	523	52.3	26.2	2.6
26.05	104800 Carbide	260.5	5370	521	26150	524	52.4	26.25	2.605
26.1	105000 Carbide	261	5380	522	26200	525	52.5	26.3	2.61
26.15	105200 Carbide	261.5	5390	523	26250	526	52.6	26.35	2.615
26.2	105400 HSS	262	5400	524	26300	527	52.7	26.4	2.62
26.25	105600 HSS	262.5	5410	525	26350	528	52.8	26.45	2.625
26.3	105800 Carbide	263	5420	526	26400	529	52.9	26.5	2.63
26.35	106000 Carbide	263.5	5430	527	26450	530	53	26.55	2.635
26.4	106200 Carbide	264	5440	528	26500	531	53.1	26.6	2.64
26.45	106400 HSS	264.5	5450	529	26550	532	53.2	26.65	2.645
26.5	106600 HSS	265	5460	530	26600	533	53.3	26.7	2.65
26.55	106800 Carbide	265.5	5470	531	26650	534	53.4	26.75	2.655
26.6	107000 Carbide	266	5480	532	26700	535	53.5	26.8	2.66
26.65	107200 Carbide	266.5	5490	533	26750	536	53.6	26.85	2.665
26.7	107400 HSS	267	5500	534	26800	537	53.7	26.9	2.67
26.75	107600 HSS	267.5	5510	535	26850	538	53.8	26.95	2.675
26.8	107800 Carbide	268	5520	536	26900	539	53.9	27	2.68
26.85	108000 Carbide	268.5	5530	537	26950	540	54	27.05	2.685
26.9	108200 Carbide	269	5540	538	27000	541	54.1	27.1	2.69
26.95	108400 HSS	269.5	5550	539	27050	542	54.2	27.15	2.695
27	108600 HSS	270	5560	540	27100	543	54.3	27.2	2.7
27.05	108800 Carbide	270.5	5570	541	27150	544	54.4	27.25	2.705
27.1	109000 Carbide	271	5580	542	27200	545	54.5	27.3	2.71
27.15	109200 Carbide	271.5	5590	543	27250	546	54.6	27.35	2.715
27.2	109400 HSS	272	5600	544	27300	547	54.7	27.4	2.72
27.25	109600 HSS	272.5	5610	545	27350	548	54.8	27.45	2.725
27.3	109800 Carbide	273	5620	546	27400	549	54.9	27.5	2.73
27.35	110000 Carbide	273.5	5630	547	27450	550	55	27.55	2.735
27.4	110200 Carbide	274	5640	548	27500	551	55.1	27.6	2.74
27.45	110400 HSS	274.5	5650	549	27550	552	55.2	27.65	2.745
27.5	110600 HSS	275	5660	550	27600	553	55.3	27.7	2.75
27.55	110800 Carbide	275.5	5670	551	27650	554	55.4	27.75	2.755
27.6	111000 Carbide	276	5680	552	27700	555	55.5	27.8	2.76
27.65	111200 Carbide	276.5	5690	553	27750	556	55.6	27.85	2.765
27.7	111400 HSS	277	5700	554	27800	557	55.7	27.9	2.77
27.75	111600 HSS	277.5	5710	555	27850	558	55.8	27.95	2.775
27.8	111800 Carbide	278	5720	556	27900	559	55.9	28	2.78
27.85	112000 Carbide	278.5	5730	557	27950	560	56	28.05	2.785
27.9	112200 Carbide	279	5740	558	28000	561	56.1	28.1	2.79
27.95	112400 HSS	279.5	5750	559	28050	562	56.2	28.15	2.795
28	112600 HSS	280	5760	560	28100	563	56.3	28.2	2.8
28.05	112800 Carbide	280.5	5770	561	28150	564	56.4	28.25	2.805
28.1	113000 Carbide	281	5780	562	28200	565	56.5	28.3	2.81
28.15	113200 Carbide	281.5	5790	563	28250	566	56.6	28.35	2.815
28.2	113400 HSS	282	5800	564	28300	567	56.7	28.4	2.82

28.25	113600 HSS	282.5	5810	565	28350	568	56.8	28.45	2.825
28.3	113800 Carbide	283	5820	566	28400	569	56.9	28.5	2.83
28.35	114000 Carbide	283.5	5830	567	28450	570	57	28.55	2.835
28.4	114200 Carbide	284	5840	568	28500	571	57.1	28.6	2.84
28.45	114400 HSS	284.5	5850	569	28550	572	57.2	28.65	2.845
28.5	114600 HSS	285	5860	570	28600	573	57.3	28.7	2.85
28.55	114800 Carbide	285.5	5870	571	28650	574	57.4	28.75	2.855
28.6	115000 Carbide	286	5880	572	28700	575	57.5	28.8	2.86
28.65	115200 Carbide	286.5	5890	573	28750	576	57.6	28.85	2.865
28.7	115400 HSS	287	5900	574	28800	577	57.7	28.9	2.87
28.75	115600 HSS	287.5	5910	575	28850	578	57.8	28.95	2.875
28.8	115800 Carbide	288	5920	576	28900	579	57.9	29	2.88
28.85	116000 Carbide	288.5	5930	577	28950	580	58	29.05	2.885
28.9	116200 Carbide	289	5940	578	29000	581	58.1	29.1	2.89
28.95	116400 HSS	289.5	5950	579	29050	582	58.2	29.15	2.895
29	116600 HSS	290	5960	580	29100	583	58.3	29.2	2.9
29.05	116800 Carbide	290.5	5970	581	29150	584	58.4	29.25	2.905
29.1	117000 Carbide	291	5980	582	29200	585	58.5	29.3	2.91
29.15	117200 Carbide	291.5	5990	583	29250	586	58.6	29.35	2.915
29.2	117400 HSS	292	6000	584	29300	587	58.7	29.4	2.92
29.25	117600 HSS	292.5	6010	585	29350	588	58.8	29.45	2.925
29.3	117800 Carbide	293	6020	586	29400	589	58.9	29.5	2.93
29.35	118000 Carbide	293.5	6030	587	29450	590	59	29.55	2.935
29.4	118200 Carbide	294	6040	588	29500	591	59.1	29.6	2.94
29.45	118400 HSS	294.5	6050	589	29550	592	59.2	29.65	2.945
29.5	118600 HSS	295	6060	590	29600	593	59.3	29.7	2.95
29.55	118800 Carbide	295.5	6070	591	29650	594	59.4	29.75	2.955
29.6	119000 Carbide	296	6080	592	29700	595	59.5	29.8	2.96
29.65	119200 Carbide	296.5	6090	593	29750	596	59.6	29.85	2.965
29.7	119400 HSS	297	6100	594	29800	597	59.7	29.9	2.97
29.75	119600 HSS	297.5	6110	595	29850	598	59.8	29.95	2.975
29.8	119800 Carbide	298	6120	596	29900	599	59.9	30	2.98
29.85	120000 Carbide	298.5	6130	597	29950	600	60	30.05	2.985
29.9	120200 Carbide	299	6140	598	30000	601	60.1	30.1	2.99
29.95	120400 HSS	299.5	6150	599	30050	602	60.2	30.15	2.995
30	120600 HSS	300	6160	600	30100	603	60.3	30.2	3
30.05	120800 Carbide	300.5	6170	601	30150	604	60.4	30.25	3.005
30.1	121000 Carbide	301	6180	602	30200	605	60.5	30.3	3.01
30.15	121200 Carbide	301.5	6190	603	30250	606	60.6	30.35	3.015
30.2	121400 HSS	302	6200	604	30300	607	60.7	30.4	3.02
30.25	121600 HSS	302.5	6210	605	30350	608	60.8	30.45	3.025
30.3	121800 Carbide	303	6220	606	30400	609	60.9	30.5	3.03
30.35	122000 Carbide	303.5	6230	607	30450	610	61	30.55	3.035
30.4	122200 Carbide	304	6240	608	30500	611	61.1	30.6	3.04
30.45	122400 HSS	304.5	6250	609	30550	612	61.2	30.65	3.045
30.5	122600 HSS	305	6260	610	30600	613	61.3	30.7	3.05
30.55	122800 Carbide	305.5	6270	611	30650	614	61.4	30.75	3.055

30.6	123000 Carbide	306	6280	612	30700	615	61.5	30.8	3.06
30.65	123200 Carbide	306.5	6290	613	30750	616	61.6	30.85	3.065
30.7	123400 HSS	307	6300	614	30800	617	61.7	30.9	3.07
30.75	123600 HSS	307.5	6310	615	30850	618	61.8	30.95	3.075
30.8	123800 Carbide	308	6320	616	30900	619	61.9	31	3.08
30.85	124000 Carbide	308.5	6330	617	30950	620	62	31.05	3.085
30.9	124200 Carbide	309	6340	618	31000	621	62.1	31.1	3.09
30.95	124400 HSS	309.5	6350	619	31050	622	62.2	31.15	3.095
31	124600 HSS	310	6360	620	31100	623	62.3	31.2	3.1
31.05	124800 Carbide	310.5	6370	621	31150	624	62.4	31.25	3.105
31.1	125000 Carbide	311	6380	622	31200	625	62.5	31.3	3.11
31.15	125200 Carbide	311.5	6390	623	31250	626	62.6	31.35	3.115
31.2	125400 HSS	312	6400	624	31300	627	62.7	31.4	3.12
31.25	125600 HSS	312.5	6410	625	31350	628	62.8	31.45	3.125
31.3	125800 Carbide	313	6420	626	31400	629	62.9	31.5	3.13
31.35	126000 Carbide	313.5	6430	627	31450	630	63	31.55	3.135
31.4	126200 Carbide	314	6440	628	31500	631	63.1	31.6	3.14
31.45	126400 HSS	314.5	6450	629	31550	632	63.2	31.65	3.145
31.5	126600 HSS	315	6460	630	31600	633	63.3	31.7	3.15
31.55	126800 Carbide	315.5	6470	631	31650	634	63.4	31.75	3.155
31.6	127000 Carbide	316	6480	632	31700	635	63.5	31.8	3.16
31.65	127200 Carbide	316.5	6490	633	31750	636	63.6	31.85	3.165
31.7	127400 HSS	317	6500	634	31800	637	63.7	31.9	3.17
31.75	127600 HSS	317.5	6510	635	31850	638	63.8	31.95	3.175
31.8	127800 Carbide	318	6520	636	31900	639	63.9	32	3.18
31.85	128000 Carbide	318.5	6530	637	31950	640	64	32.05	3.185
31.9	128200 Carbide	319	6540	638	32000	641	64.1	32.1	3.19
31.95	128400 HSS	319.5	6550	639	32050	642	64.2	32.15	3.195
32	128600 HSS	320	6560	640	32100	643	64.3	32.2	3.2
32.05	128800 Carbide	320.5	6570	641	32150	644	64.4	32.25	3.205
32.1	129000 Carbide	321	6580	642	32200	645	64.5	32.3	3.21
32.15	129200 Carbide	321.5	6590	643	32250	646	64.6	32.35	3.215
32.2	129400 HSS	322	6600	644	32300	647	64.7	32.4	3.22
32.25	129600 HSS	322.5	6610	645	32350	648	64.8	32.45	3.225
32.3	129800 Carbide	323	6620	646	32400	649	64.9	32.5	3.23
32.35	130000 Carbide	323.5	6630	647	32450	650	65	32.55	3.235
32.4	130200 Carbide	324	6640	648	32500	651	65.1	32.6	3.24
32.45	130400 HSS	324.5	6650	649	32550	652	65.2	32.65	3.245
32.5	130600 HSS	325	6660	650	32600	653	65.3	32.7	3.25
32.55	130800 Carbide	325.5	6670	651	32650	654	65.4	32.75	3.255
32.6	131000 Carbide	326	6680	652	32700	655	65.5	32.8	3.26
32.65	131200 Carbide	326.5	6690	653	32750	656	65.6	32.85	3.265
32.7	131400 HSS	327	6700	654	32800	657	65.7	32.9	3.27
32.75	131600 HSS	327.5	6710	655	32850	658	65.8	32.95	3.275
32.8	131800 Carbide	328	6720	656	32900	659	65.9	33	3.28
32.85	132000 Carbide	328.5	6730	657	32950	660	66	33.05	3.285
32.9	132200 Carbide	329	6740	658	33000	661	66.1	33.1	3.29

32.95	132400 HSS	329.5	6750	659	33050	662	66.2	33.15	3.295
33	132600 HSS	330	6760	660	33100	663	66.3	33.2	3.3
33.05	132800 Carbide	330.5	6770	661	33150	664	66.4	33.25	3.305
33.1	133000 Carbide	331	6780	662	33200	665	66.5	33.3	3.31
33.15	133200 Carbide	331.5	6790	663	33250	666	66.6	33.35	3.315
33.2	133400 HSS	332	6800	664	33300	667	66.7	33.4	3.32
33.25	133600 HSS	332.5	6810	665	33350	668	66.8	33.45	3.325
33.3	133800 Carbide	333	6820	666	33400	669	66.9	33.5	3.33
33.35	134000 Carbide	333.5	6830	667	33450	670	67	33.55	3.335
33.4	134200 Carbide	334	6840	668	33500	671	67.1	33.6	3.34
33.45	134400 HSS	334.5	6850	669	33550	672	67.2	33.65	3.345
33.5	134600 HSS	335	6860	670	33600	673	67.3	33.7	3.35
33.55	134800 Carbide	335.5	6870	671	33650	674	67.4	33.75	3.355
33.6	135000 Carbide	336	6880	672	33700	675	67.5	33.8	3.36
33.65	135200 Carbide	336.5	6890	673	33750	676	67.6	33.85	3.365
33.7	135400 HSS	337	6900	674	33800	677	67.7	33.9	3.37
33.75	135600 HSS	337.5	6910	675	33850	678	67.8	33.95	3.375
33.8	135800 Carbide	338	6920	676	33900	679	67.9	34	3.38
33.85	136000 Carbide	338.5	6930	677	33950	680	68	34.05	3.385
33.9	136200 Carbide	339	6940	678	34000	681	68.1	34.1	3.39
33.95	136400 HSS	339.5	6950	679	34050	682	68.2	34.15	3.395
34	136600 HSS	340	6960	680	34100	683	68.3	34.2	3.4
34.05	136800 Carbide	340.5	6970	681	34150	684	68.4	34.25	3.405
34.1	137000 Carbide	341	6980	682	34200	685	68.5	34.3	3.41
34.15	137200 Carbide	341.5	6990	683	34250	686	68.6	34.35	3.415
34.2	137400 HSS	342	7000	684	34300	687	68.7	34.4	3.42
34.25	137600 HSS	342.5	7010	685	34350	688	68.8	34.45	3.425
34.3	137800 Carbide	343	7020	686	34400	689	68.9	34.5	3.43
34.35	138000 Carbide	343.5	7030	687	34450	690	69	34.55	3.435
34.4	138200 Carbide	344	7040	688	34500	691	69.1	34.6	3.44
34.45	138400 HSS	344.5	7050	689	34550	692	69.2	34.65	3.445
34.5	138600 HSS	345	7060	690	34600	693	69.3	34.7	3.45
34.55	138800 Carbide	345.5	7070	691	34650	694	69.4	34.75	3.455
34.6	139000 Carbide	346	7080	692	34700	695	69.5	34.8	3.46
34.65	139200 Carbide	346.5	7090	693	34750	696	69.6	34.85	3.465
34.7	139400 HSS	347	7100	694	34800	697	69.7	34.9	3.47
34.75	139600 HSS	347.5	7110	695	34850	698	69.8	34.95	3.475
34.8	139800 Carbide	348	7120	696	34900	699	69.9	35	3.48
34.85	140000 Carbide	348.5	7130	697	34950	700	70	35.05	3.485
34.9	140200 Carbide	349	7140	698	35000	701	70.1	35.1	3.49
34.95	140400 HSS	349.5	7150	699	35050	702	70.2	35.15	3.495
35	140600 HSS	350	7160	700	35100	703	70.3	35.2	3.5
35.05	140800 Carbide	350.5	7170	701	35150	704	70.4	35.25	3.505
35.1	141000 Carbide	351	7180	702	35200	705	70.5	35.3	3.51
35.15	141200 Carbide	351.5	7190	703	35250	706	70.6	35.35	3.515
35.2	141400 HSS	352	7200	704	35300	707	70.7	35.4	3.52
35.25	141600 HSS	352.5	7210	705	35350	708	70.8	35.45	3.525

35.3	141800 Carbide	353	7220	706	35400	709	70.9	35.5	3.53
35.35	142000 Carbide	353.5	7230	707	35450	710	71	35.55	3.535
35.4	142200 Carbide	354	7240	708	35500	711	71.1	35.6	3.54
35.45	142400 HSS	354.5	7250	709	35550	712	71.2	35.65	3.545
35.5	142600 HSS	355	7260	710	35600	713	71.3	35.7	3.55
35.55	142800 Carbide	355.5	7270	711	35650	714	71.4	35.75	3.555
35.6	143000 Carbide	356	7280	712	35700	715	71.5	35.8	3.56
35.65	143200 Carbide	356.5	7290	713	35750	716	71.6	35.85	3.565
35.7	143400 HSS	357	7300	714	35800	717	71.7	35.9	3.57
35.75	143600 HSS	357.5	7310	715	35850	718	71.8	35.95	3.575
35.8	143800 Carbide	358	7320	716	35900	719	71.9	36	3.58
35.85	144000 Carbide	358.5	7330	717	35950	720	72	36.05	3.585
35.9	144200 Carbide	359	7340	718	36000	721	72.1	36.1	3.59
35.95	144400 HSS	359.5	7350	719	36050	722	72.2	36.15	3.595
36	144600 HSS	360	7360	720	36100	723	72.3	36.2	3.6
36.05	144800 Carbide	360.5	7370	721	36150	724	72.4	36.25	3.605
36.1	145000 Carbide	361	7380	722	36200	725	72.5	36.3	3.61
36.15	145200 Carbide	361.5	7390	723	36250	726	72.6	36.35	3.615
36.2	145400 HSS	362	7400	724	36300	727	72.7	36.4	3.62
36.25	145600 HSS	362.5	7410	725	36350	728	72.8	36.45	3.625
36.3	145800 Carbide	363	7420	726	36400	729	72.9	36.5	3.63
36.35	146000 Carbide	363.5	7430	727	36450	730	73	36.55	3.635
36.4	146200 Carbide	364	7440	728	36500	731	73.1	36.6	3.64
36.45	146400 HSS	364.5	7450	729	36550	732	73.2	36.65	3.645
36.5	146600 HSS	365	7460	730	36600	733	73.3	36.7	3.65
36.55	146800 Carbide	365.5	7470	731	36650	734	73.4	36.75	3.655
36.6	147000 Carbide	366	7480	732	36700	735	73.5	36.8	3.66
36.65	147200 Carbide	366.5	7490	733	36750	736	73.6	36.85	3.665
36.7	147400 HSS	367	7500	734	36800	737	73.7	36.9	3.67
36.75	147600 HSS	367.5	7510	735	36850	738	73.8	36.95	3.675
36.8	147800 Carbide	368	7520	736	36900	739	73.9	37	3.68
36.85	148000 Carbide	368.5	7530	737	36950	740	74	37.05	3.685
36.9	148200 Carbide	369	7540	738	37000	741	74.1	37.1	3.69
36.95	148400 HSS	369.5	7550	739	37050	742	74.2	37.15	3.695
37	148600 HSS	370	7560	740	37100	743	74.3	37.2	3.7
37.05	148800 Carbide	370.5	7570	741	37150	744	74.4	37.25	3.705
37.1	149000 Carbide	371	7580	742	37200	745	74.5	37.3	3.71
37.15	149200 Carbide	371.5	7590	743	37250	746	74.6	37.35	3.715
37.2	149400 HSS	372	7600	744	37300	747	74.7	37.4	3.72
37.25	149600 HSS	372.5	7610	745	37350	748	74.8	37.45	3.725
37.3	149800 Carbide	373	7620	746	37400	749	74.9	37.5	3.73
37.35	150000 Carbide	373.5	7630	747	37450	750	75	37.55	3.735
37.4	150200 Carbide	374	7640	748	37500	751	75.1	37.6	3.74
37.45	150400 HSS	374.5	7650	749	37550	752	75.2	37.65	3.745
37.5	150600 HSS	375	7660	750	37600	753	75.3	37.7	3.75
37.55	150800 Carbide	375.5	7670	751	37650	754	75.4	37.75	3.755
37.6	151000 Carbide	376	7680	752	37700	755	75.5	37.8	3.76

37.65	151200 Carbide	376.5	7690	753	37750	756	75.6	37.85	3.765
37.7	151400 HSS	377	7700	754	37800	757	75.7	37.9	3.77
37.75	151600 HSS	377.5	7710	755	37850	758	75.8	37.95	3.775
37.8	151800 Carbide	378	7720	756	37900	759	75.9	38	3.78
37.85	152000 Carbide	378.5	7730	757	37950	760	76	38.05	3.785
37.9	152200 Carbide	379	7740	758	38000	761	76.1	38.1	3.79
37.95	152400 HSS	379.5	7750	759	38050	762	76.2	38.15	3.795
38	152600 HSS	380	7760	760	38100	763	76.3	38.2	3.8
38.05	152800 Carbide	380.5	7770	761	38150	764	76.4	38.25	3.805
38.1	153000 Carbide	381	7780	762	38200	765	76.5	38.3	3.81
38.15	153200 Carbide	381.5	7790	763	38250	766	76.6	38.35	3.815
38.2	153400 HSS	382	7800	764	38300	767	76.7	38.4	3.82
38.25	153600 HSS	382.5	7810	765	38350	768	76.8	38.45	3.825
38.3	153800 Carbide	383	7820	766	38400	769	76.9	38.5	3.83
38.35	154000 Carbide	383.5	7830	767	38450	770	77	38.55	3.835
38.4	154200 Carbide	384	7840	768	38500	771	77.1	38.6	3.84
38.45	154400 HSS	384.5	7850	769	38550	772	77.2	38.65	3.845
38.5	154600 HSS	385	7860	770	38600	773	77.3	38.7	3.85
38.55	154800 Carbide	385.5	7870	771	38650	774	77.4	38.75	3.855
38.6	155000 Carbide	386	7880	772	38700	775	77.5	38.8	3.86
38.65	155200 Carbide	386.5	7890	773	38750	776	77.6	38.85	3.865
38.7	155400 HSS	387	7900	774	38800	777	77.7	38.9	3.87
38.75	155600 HSS	387.5	7910	775	38850	778	77.8	38.95	3.875
38.8	155800 Carbide	388	7920	776	38900	779	77.9	39	3.88
38.85	156000 Carbide	388.5	7930	777	38950	780	78	39.05	3.885
38.9	156200 Carbide	389	7940	778	39000	781	78.1	39.1	3.89
38.95	156400 HSS	389.5	7950	779	39050	782	78.2	39.15	3.895
39	156600 HSS	390	7960	780	39100	783	78.3	39.2	3.9
39.05	156800 Carbide	390.5	7970	781	39150	784	78.4	39.25	3.905
39.1	157000 Carbide	391	7980	782	39200	785	78.5	39.3	3.91
39.15	157200 Carbide	391.5	7990	783	39250	786	78.6	39.35	3.915
39.2	157400 HSS	392	8000	784	39300	787	78.7	39.4	3.92
39.25	157600 HSS	392.5	8010	785	39350	788	78.8	39.45	3.925
39.3	157800 Carbide	393	8020	786	39400	789	78.9	39.5	3.93
39.35	158000 Carbide	393.5	8030	787	39450	790	79	39.55	3.935
39.4	158200 Carbide	394	8040	788	39500	791	79.1	39.6	3.94
39.45	158400 HSS	394.5	8050	789	39550	792	79.2	39.65	3.945
39.5	158600 HSS	395	8060	790	39600	793	79.3	39.7	3.95
39.55	158800 Carbide	395.5	8070	791	39650	794	79.4	39.75	3.955
39.6	159000 Carbide	396	8080	792	39700	795	79.5	39.8	3.96
39.65	159200 Carbide	396.5	8090	793	39750	796	79.6	39.85	3.965
39.7	159400 HSS	397	8100	794	39800	797	79.7	39.9	3.97
39.75	159600 HSS	397.5	8110	795	39850	798	79.8	39.95	3.975
39.8	159800 Carbide	398	8120	796	39900	799	79.9	40	3.98
39.85	160000 Carbide	398.5	8130	797	39950	800	80	40.05	3.985
39.9	160200 Carbide	399	8140	798	40000	801	80.1	40.1	3.99
39.95	160400 HSS	399.5	8150	799	40050	802	80.2	40.15	3.995

40	160600 HSS	400	8160	800	40100	803	80.3	40.2	4
40.05	160800 Carbide	400.5	8170	801	40150	804	80.4	40.25	4.005
40.1	161000 Carbide	401	8180	802	40200	805	80.5	40.3	4.01
40.15	161200 Carbide	401.5	8190	803	40250	806	80.6	40.35	4.015
40.2	161400 HSS	402	8200	804	40300	807	80.7	40.4	4.02
40.25	161600 HSS	402.5	8210	805	40350	808	80.8	40.45	4.025
40.3	161800 Carbide	403	8220	806	40400	809	80.9	40.5	4.03
40.35	162000 Carbide	403.5	8230	807	40450	810	81	40.55	4.035
40.4	162200 Carbide	404	8240	808	40500	811	81.1	40.6	4.04
40.45	162400 HSS	404.5	8250	809	40550	812	81.2	40.65	4.045
40.5	162600 HSS	405	8260	810	40600	813	81.3	40.7	4.05
40.55	162800 Carbide	405.5	8270	811	40650	814	81.4	40.75	4.055
40.6	163000 Carbide	406	8280	812	40700	815	81.5	40.8	4.06
40.65	163200 Carbide	406.5	8290	813	40750	816	81.6	40.85	4.065
40.7	163400 HSS	407	8300	814	40800	817	81.7	40.9	4.07
40.75	163600 HSS	407.5	8310	815	40850	818	81.8	40.95	4.075
40.8	163800 Carbide	408	8320	816	40900	819	81.9	41	4.08
40.85	164000 Carbide	408.5	8330	817	40950	820	82	41.05	4.085
40.9	164200 Carbide	409	8340	818	41000	821	82.1	41.1	4.09
40.95	164400 HSS	409.5	8350	819	41050	822	82.2	41.15	4.095
41	164600 HSS	410	8360	820	41100	823	82.3	41.2	4.1
41.05	164800 Carbide	410.5	8370	821	41150	824	82.4	41.25	4.105
41.1	165000 Carbide	411	8380	822	41200	825	82.5	41.3	4.11
41.15	165200 Carbide	411.5	8390	823	41250	826	82.6	41.35	4.115
41.2	165400 HSS	412	8400	824	41300	827	82.7	41.4	4.12
41.25	165600 HSS	412.5	8410	825	41350	828	82.8	41.45	4.125
41.3	165800 Carbide	413	8420	826	41400	829	82.9	41.5	4.13
41.35	166000 Carbide	413.5	8430	827	41450	830	83	41.55	4.135
41.4	166200 Carbide	414	8440	828	41500	831	83.1	41.6	4.14
41.45	166400 HSS	414.5	8450	829	41550	832	83.2	41.65	4.145
41.5	166600 HSS	415	8460	830	41600	833	83.3	41.7	4.15
41.55	166800 Carbide	415.5	8470	831	41650	834	83.4	41.75	4.155
41.6	167000 Carbide	416	8480	832	41700	835	83.5	41.8	4.16
41.65	167200 Carbide	416.5	8490	833	41750	836	83.6	41.85	4.165
41.7	167400 HSS	417	8500	834	41800	837	83.7	41.9	4.17
41.75	167600 HSS	417.5	8510	835	41850	838	83.8	41.95	4.175
41.8	167800 Carbide	418	8520	836	41900	839	83.9	42	4.18
41.85	168000 Carbide	418.5	8530	837	41950	840	84	42.05	4.185
41.9	168200 Carbide	419	8540	838	42000	841	84.1	42.1	4.19
41.95	168400 HSS	419.5	8550	839	42050	842	84.2	42.15	4.195
42	168600 HSS	420	8560	840	42100	843	84.3	42.2	4.2
42.05	168800 Carbide	420.5	8570	841	42150	844	84.4	42.25	4.205
42.1	169000 Carbide	421	8580	842	42200	845	84.5	42.3	4.21
42.15	169200 Carbide	421.5	8590	843	42250	846	84.6	42.35	4.215
42.2	169400 HSS	422	8600	844	42300	847	84.7	42.4	4.22
42.25	169600 HSS	422.5	8610	845	42350	848	84.8	42.45	4.225
42.3	169800 Carbide	423	8620	846	42400	849	84.9	42.5	4.23

42.35	170000 Carbide	423.5	8630	847	42450	850	85	42.55	4.235
42.4	170200 Carbide	424	8640	848	42500	851	85.1	42.6	4.24
42.45	170400 HSS	424.5	8650	849	42550	852	85.2	42.65	4.245
42.5	170600 HSS	425	8660	850	42600	853	85.3	42.7	4.25
42.55	170800 Carbide	425.5	8670	851	42650	854	85.4	42.75	4.255
42.6	171000 Carbide	426	8680	852	42700	855	85.5	42.8	4.26
42.65	171200 Carbide	426.5	8690	853	42750	856	85.6	42.85	4.265
42.7	171400 HSS	427	8700	854	42800	857	85.7	42.9	4.27
42.75	171600 HSS	427.5	8710	855	42850	858	85.8	42.95	4.275
42.8	171800 Carbide	428	8720	856	42900	859	85.9	43	4.28
42.85	172000 Carbide	428.5	8730	857	42950	860	86	43.05	4.285
42.9	172200 Carbide	429	8740	858	43000	861	86.1	43.1	4.29
42.95	172400 HSS	429.5	8750	859	43050	862	86.2	43.15	4.295
43	172600 HSS	430	8760	860	43100	863	86.3	43.2	4.3
43.05	172800 Carbide	430.5	8770	861	43150	864	86.4	43.25	4.305
43.1	173000 Carbide	431	8780	862	43200	865	86.5	43.3	4.31
43.15	173200 Carbide	431.5	8790	863	43250	866	86.6	43.35	4.315
43.2	173400 HSS	432	8800	864	43300	867	86.7	43.4	4.32
43.25	173600 HSS	432.5	8810	865	43350	868	86.8	43.45	4.325
43.3	173800 Carbide	433	8820	866	43400	869	86.9	43.5	4.33
43.35	174000 Carbide	433.5	8830	867	43450	870	87	43.55	4.335
43.4	174200 Carbide	434	8840	868	43500	871	87.1	43.6	4.34
43.45	174400 HSS	434.5	8850	869	43550	872	87.2	43.65	4.345
43.5	174600 HSS	435	8860	870	43600	873	87.3	43.7	4.35
43.55	174800 Carbide	435.5	8870	871	43650	874	87.4	43.75	4.355
43.6	175000 Carbide	436	8880	872	43700	875	87.5	43.8	4.36
43.65	175200 Carbide	436.5	8890	873	43750	876	87.6	43.85	4.365
43.7	175400 HSS	437	8900	874	43800	877	87.7	43.9	4.37
43.75	175600 HSS	437.5	8910	875	43850	878	87.8	43.95	4.375
43.8	175800 Carbide	438	8920	876	43900	879	87.9	44	4.38
43.85	176000 Carbide	438.5	8930	877	43950	880	88	44.05	4.385
43.9	176200 Carbide	439	8940	878	44000	881	88.1	44.1	4.39
43.95	176400 HSS	439.5	8950	879	44050	882	88.2	44.15	4.395
44	176600 HSS	440	8960	880	44100	883	88.3	44.2	4.4
44.05	176800 Carbide	440.5	8970	881	44150	884	88.4	44.25	4.405
44.1	177000 Carbide	441	8980	882	44200	885	88.5	44.3	4.41
44.15	177200 Carbide	441.5	8990	883	44250	886	88.6	44.35	4.415
44.2	177400 HSS	442	9000	884	44300	887	88.7	44.4	4.42
44.25	177600 HSS	442.5	9010	885	44350	888	88.8	44.45	4.425
44.3	177800 Carbide	443	9020	886	44400	889	88.9	44.5	4.43
44.35	178000 Carbide	443.5	9030	887	44450	890	89	44.55	4.435
44.4	178200 Carbide	444	9040	888	44500	891	89.1	44.6	4.44
44.45	178400 HSS	444.5	9050	889	44550	892	89.2	44.65	4.445
44.5	178600 HSS	445	9060	890	44600	893	89.3	44.7	4.45
44.55	178800 Carbide	445.5	9070	891	44650	894	89.4	44.75	4.455
44.6	179000 Carbide	446	9080	892	44700	895	89.5	44.8	4.46
44.65	179200 Carbide	446.5	9090	893	44750	896	89.6	44.85	4.465

44.7	179400 HSS	447	9100	894	44800	897	89.7	44.9	4.47
44.75	179600 HSS	447.5	9110	895	44850	898	89.8	44.95	4.475
44.8	179800 Carbide	448	9120	896	44900	899	89.9	45	4.48
44.85	180000 Carbide	448.5	9130	897	44950	900	90	45.05	4.485
44.9	180200 Carbide	449	9140	898	45000	901	90.1	45.1	4.49
44.95	180400 HSS	449.5	9150	899	45050	902	90.2	45.15	4.495
45	180600 HSS	450	9160	900	45100	903	90.3	45.2	4.5
45.05	180800 Carbide	450.5	9170	901	45150	904	90.4	45.25	4.505
45.1	181000 Carbide	451	9180	902	45200	905	90.5	45.3	4.51
45.15	181200 Carbide	451.5	9190	903	45250	906	90.6	45.35	4.515
45.2	181400 HSS	452	9200	904	45300	907	90.7	45.4	4.52
45.25	181600 HSS	452.5	9210	905	45350	908	90.8	45.45	4.525
45.3	181800 Carbide	453	9220	906	45400	909	90.9	45.5	4.53
45.35	182000 Carbide	453.5	9230	907	45450	910	91	45.55	4.535
45.4	182200 Carbide	454	9240	908	45500	911	91.1	45.6	4.54
45.45	182400 HSS	454.5	9250	909	45550	912	91.2	45.65	4.545
45.5	182600 HSS	455	9260	910	45600	913	91.3	45.7	4.55
45.55	182800 Carbide	455.5	9270	911	45650	914	91.4	45.75	4.555
45.6	183000 Carbide	456	9280	912	45700	915	91.5	45.8	4.56
45.65	183200 Carbide	456.5	9290	913	45750	916	91.6	45.85	4.565
45.7	183400 HSS	457	9300	914	45800	917	91.7	45.9	4.57
45.75	183600 HSS	457.5	9310	915	45850	918	91.8	45.95	4.575
45.8	183800 Carbide	458	9320	916	45900	919	91.9	46	4.58
45.85	184000 Carbide	458.5	9330	917	45950	920	92	46.05	4.585
45.9	184200 Carbide	459	9340	918	46000	921	92.1	46.1	4.59
45.95	184400 HSS	459.5	9350	919	46050	922	92.2	46.15	4.595
46	184600 HSS	460	9360	920	46100	923	92.3	46.2	4.6
46.05	184800 Carbide	460.5	9370	921	46150	924	92.4	46.25	4.605
46.1	185000 Carbide	461	9380	922	46200	925	92.5	46.3	4.61
46.15	185200 Carbide	461.5	9390	923	46250	926	92.6	46.35	4.615
46.2	185400 HSS	462	9400	924	46300	927	92.7	46.4	4.62
46.25	185600 HSS	462.5	9410	925	46350	928	92.8	46.45	4.625
46.3	185800 Carbide	463	9420	926	46400	929	92.9	46.5	4.63
46.35	186000 Carbide	463.5	9430	927	46450	930	93	46.55	4.635
46.4	186200 Carbide	464	9440	928	46500	931	93.1	46.6	4.64
46.45	186400 HSS	464.5	9450	929	46550	932	93.2	46.65	4.645
46.5	186600 HSS	465	9460	930	46600	933	93.3	46.7	4.65
46.55	186800 Carbide	465.5	9470	931	46650	934	93.4	46.75	4.655
46.6	187000 Carbide	466	9480	932	46700	935	93.5	46.8	4.66
46.65	187200 Carbide	466.5	9490	933	46750	936	93.6	46.85	4.665
46.7	187400 HSS	467	9500	934	46800	937	93.7	46.9	4.67
46.75	187600 HSS	467.5	9510	935	46850	938	93.8	46.95	4.675
46.8	187800 Carbide	468	9520	936	46900	939	93.9	47	4.68
46.85	188000 Carbide	468.5	9530	937	46950	940	94	47.05	4.685
46.9	188200 Carbide	469	9540	938	47000	941	94.1	47.1	4.69
46.95	188400 HSS	469.5	9550	939	47050	942	94.2	47.15	4.695
47	188600 HSS	470	9560	940	47100	943	94.3	47.2	4.7

47.05	188800 Carbide	470.5	9570	941	47150	944	94.4	47.25	4.705
47.1	189000 Carbide	471	9580	942	47200	945	94.5	47.3	4.71
47.15	189200 Carbide	471.5	9590	943	47250	946	94.6	47.35	4.715
47.2	189400 HSS	472	9600	944	47300	947	94.7	47.4	4.72
47.25	189600 HSS	472.5	9610	945	47350	948	94.8	47.45	4.725
47.3	189800 Carbide	473	9620	946	47400	949	94.9	47.5	4.73
47.35	190000 Carbide	473.5	9630	947	47450	950	95	47.55	4.735
47.4	190200 Carbide	474	9640	948	47500	951	95.1	47.6	4.74
47.45	190400 HSS	474.5	9650	949	47550	952	95.2	47.65	4.745
47.5	190600 HSS	475	9660	950	47600	953	95.3	47.7	4.75
47.55	190800 Carbide	475.5	9670	951	47650	954	95.4	47.75	4.755
47.6	191000 Carbide	476	9680	952	47700	955	95.5	47.8	4.76
47.65	191200 Carbide	476.5	9690	953	47750	956	95.6	47.85	4.765
47.7	191400 HSS	477	9700	954	47800	957	95.7	47.9	4.77
47.75	191600 HSS	477.5	9710	955	47850	958	95.8	47.95	4.775
47.8	191800 Carbide	478	9720	956	47900	959	95.9	48	4.78
47.85	192000 Carbide	478.5	9730	957	47950	960	96	48.05	4.785
47.9	192200 Carbide	479	9740	958	48000	961	96.1	48.1	4.79
47.95	192400 HSS	479.5	9750	959	48050	962	96.2	48.15	4.795
48	192600 HSS	480	9760	960	48100	963	96.3	48.2	4.8
48.05	192800 Carbide	480.5	9770	961	48150	964	96.4	48.25	4.805
48.1	193000 Carbide	481	9780	962	48200	965	96.5	48.3	4.81
48.15	193200 Carbide	481.5	9790	963	48250	966	96.6	48.35	4.815
48.2	193400 HSS	482	9800	964	48300	967	96.7	48.4	4.82
48.25	193600 HSS	482.5	9810	965	48350	968	96.8	48.45	4.825
48.3	193800 Carbide	483	9820	966	48400	969	96.9	48.5	4.83
48.35	194000 Carbide	483.5	9830	967	48450	970	97	48.55	4.835
48.4	194200 Carbide	484	9840	968	48500	971	97.1	48.6	4.84
48.45	194400 HSS	484.5	9850	969	48550	972	97.2	48.65	4.845
48.5	194600 HSS	485	9860	970	48600	973	97.3	48.7	4.85
48.55	194800 Carbide	485.5	9870	971	48650	974	97.4	48.75	4.855
48.6	195000 Carbide	486	9880	972	48700	975	97.5	48.8	4.86
48.65	195200 Carbide	486.5	9890	973	48750	976	97.6	48.85	4.865
48.7	195400 HSS	487	9900	974	48800	977	97.7	48.9	4.87
48.75	195600 HSS	487.5	9910	975	48850	978	97.8	48.95	4.875
48.8	195800 Carbide	488	9920	976	48900	979	97.9	49	4.88
48.85	196000 Carbide	488.5	9930	977	48950	980	98	49.05	4.885
48.9	196200 Carbide	489	9940	978	49000	981	98.1	49.1	4.89
48.95	196400 HSS	489.5	9950	979	49050	982	98.2	49.15	4.895
49	196600 HSS	490	9960	980	49100	983	98.3	49.2	4.9
49.05	196800 Carbide	490.5	9970	981	49150	984	98.4	49.25	4.905
49.1	197000 Carbide	491	9980	982	49200	985	98.5	49.3	4.91
49.15	197200 Carbide	491.5	9990	983	49250	986	98.6	49.35	4.915
49.2	197400 HSS	492	10000	984	49300	987	98.7	49.4	4.92
49.25	197600 HSS	492.5	10010	985	49350	988	98.8	49.45	4.925
49.3	197800 Carbide	493	10020	986	49400	989	98.9	49.5	4.93
49.35	198000 Carbide	493.5	10030	987	49450	990	99	49.55	4.935

49.4	198200 Carbide	494	10040	988	49500	991	99.1	49.6	4.94
49.45	198400 HSS	494.5	10050	989	49550	992	99.2	49.65	4.945
49.5	198600 HSS	495	10060	990	49600	993	99.3	49.7	4.95
49.55	198800 Carbide	495.5	10070	991	49650	994	99.4	49.75	4.955
49.6	199000 Carbide	496	10080	992	49700	995	99.5	49.8	4.96
49.65	199200 Carbide	496.5	10090	993	49750	996	99.6	49.85	4.965
49.7	199400 HSS	497	10100	994	49800	997	99.7	49.9	4.97
49.75	199600 HSS	497.5	10110	995	49850	998	99.8	49.95	4.975
49.8	199800 Carbide	498	10120	996	49900	999	99.9	50	4.98
49.85	200000 Carbide	498.5	10130	997	49950	1000	100	50.05	4.985
49.9	200200 Carbide	499	10140	998	50000	1001	100.1	50.1	4.99
49.95	200400 HSS	499.5	10150	999	50050	1002	100.2	50.15	4.995
50	200600 HSS	500	10160	1000	50100	1003	100.3	50.2	5
50.05	200800 Carbide	500.5	10170	1001	50150	1004	100.4	50.25	5.005